

Solution Requirements

Date	10/07/2026
Name	Karuru Arjun
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements that define what the Voice-Based Concept Understanding Analyser (VBCUA) should do and how effectively it should perform.

Functional Requirements (FR)

Functional requirements describe the features and operations of the VBCUA system, including audio upload and recording, speech-to-text conversion, semantic similarity evaluation, speech fluency analysis, AI-based scoring, report generation, and secure storage. These requirements define what the system should do.

Non-Functional Requirements (NFR)

Non-functional requirements define the quality attributes of the VBCUA system, including performance, accuracy, reliability, security, scalability, usability, and maintainability. These describe how well the system should perform

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Authentication	Users can securely access the application and upload or record voice explanations.	High
2	Audio Input Module	The system should allow users to upload audio files or record speech directly from the application.	High
3	Speech-to-Text Processing	The application should convert spoken audio into text using the OpenAI Whisper model	High
4	Semantic Analysis	The system should compare the transcribed explanation with the reference concept using Sentence-BERT and generate a semantic	High

		similarity score.	
5	Speech Fluency Analysis	The application should analyze speech duration, pause ratio, RMS energy, and filler words using Librosa.	High
6	Result Generation	The system should display transcript, semantic score, fluency metrics, overall performance score, and AI-generated feedback.	High
7	PDF Report Generation	Users should be able to download a detailed evaluation report containing transcript, scores, waveform, and feedback.	Medium
8	Data Management	The application should securely store evaluation data and support future model improvements.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should quickly process uploaded audio and generate evaluation results.	Report generated within 5–10 seconds for normal audio files.
2	Accuracy	Speech transcription and semantic evaluation should produce reliable results.	90%+ transcription accuracy for clear audio.
3	Security & Data Privacy	Uploaded audio files and reports should be securely stored and protected.	Secure file handling and protected user data
4	Reliability & Availability	The application should provide uninterrupted service with minimal downtime.	99% system availability
5	Usability & Accessibility	The application should provide a simple, user-friendly interface for students and educators.	Easy navigation and responsive dashboard
6	Other	The system should be maintainable and allow easy model updates with new datasets.	Supports model retraining and future enhancements